Poisson regression

Ciaran Evans

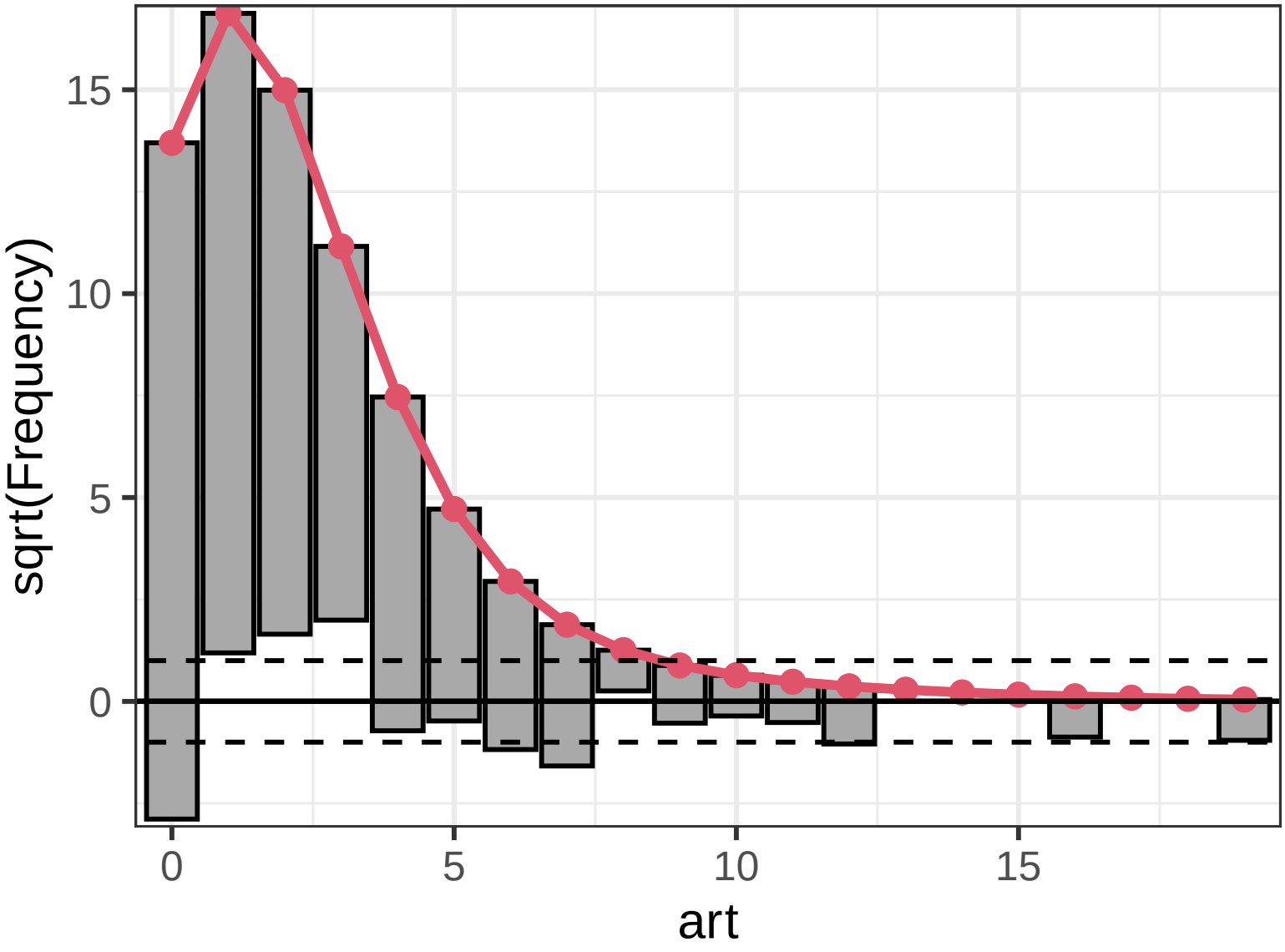
Activity

Use the provided packet to answer the following questions:

- Does the Poisson regression model appear to be a good fit?
- If not, which piece of the model do you think is mis-specified?

Rootograms: an overall assessment of model fit

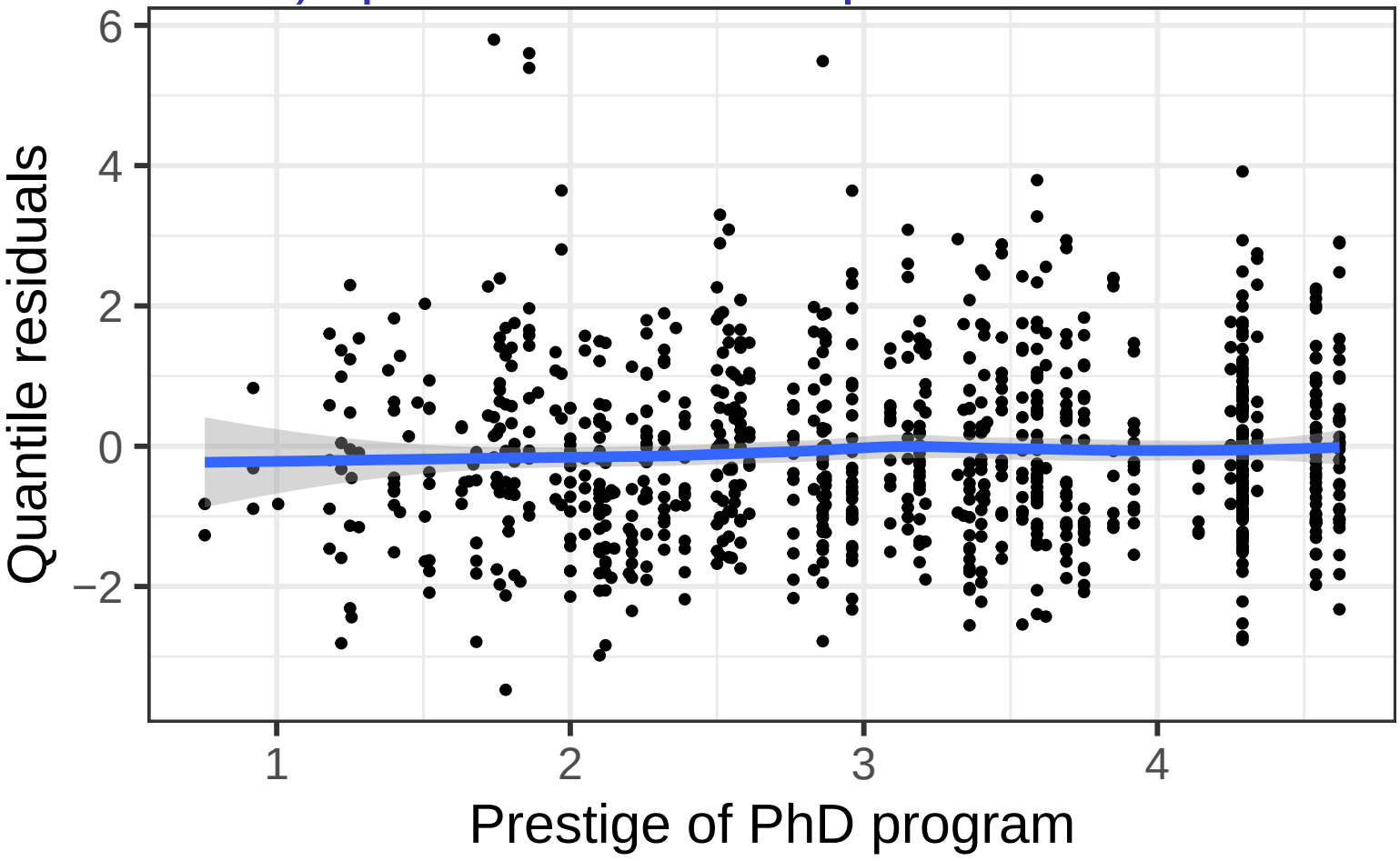
```
topmodels::rootogram(m_art)
```



Rootograms: an overall assessment of model fit

- horizontal axis: possible values for the response variable
- vertical axis (square root scale): predicted count for the model (points) and *observed* counts from the data (bars)
- Assessing whether predicted counts match observed counts
- Lack of fit could be due to:
  - Wrong choice of distribution
  - Additional explanatory variables need to be added to model
  - Issues with shape assumption

(Randomized) quantile residual plots



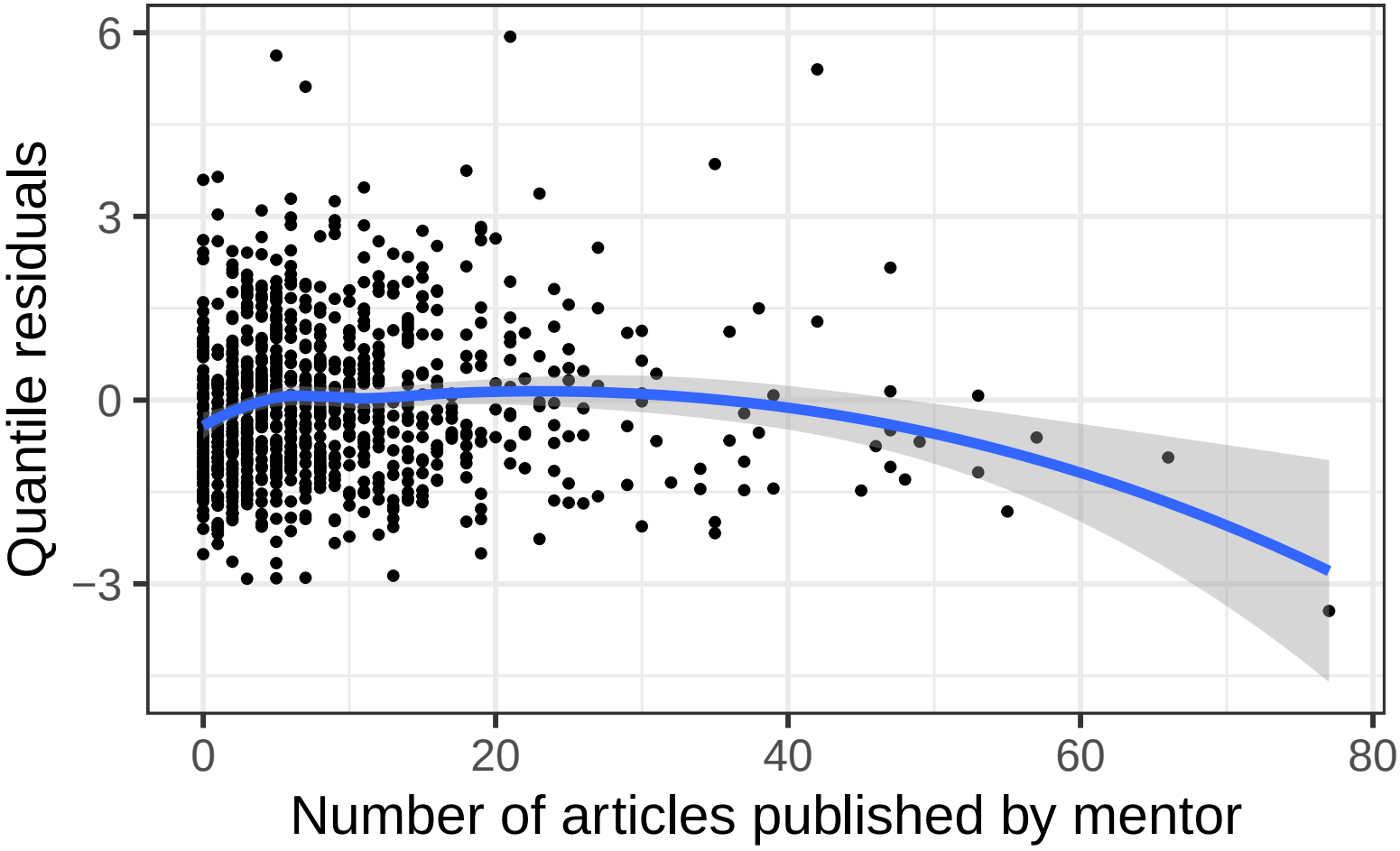
What I am looking for in a quantile residual plot?

Examining influential points

Cook's distance:

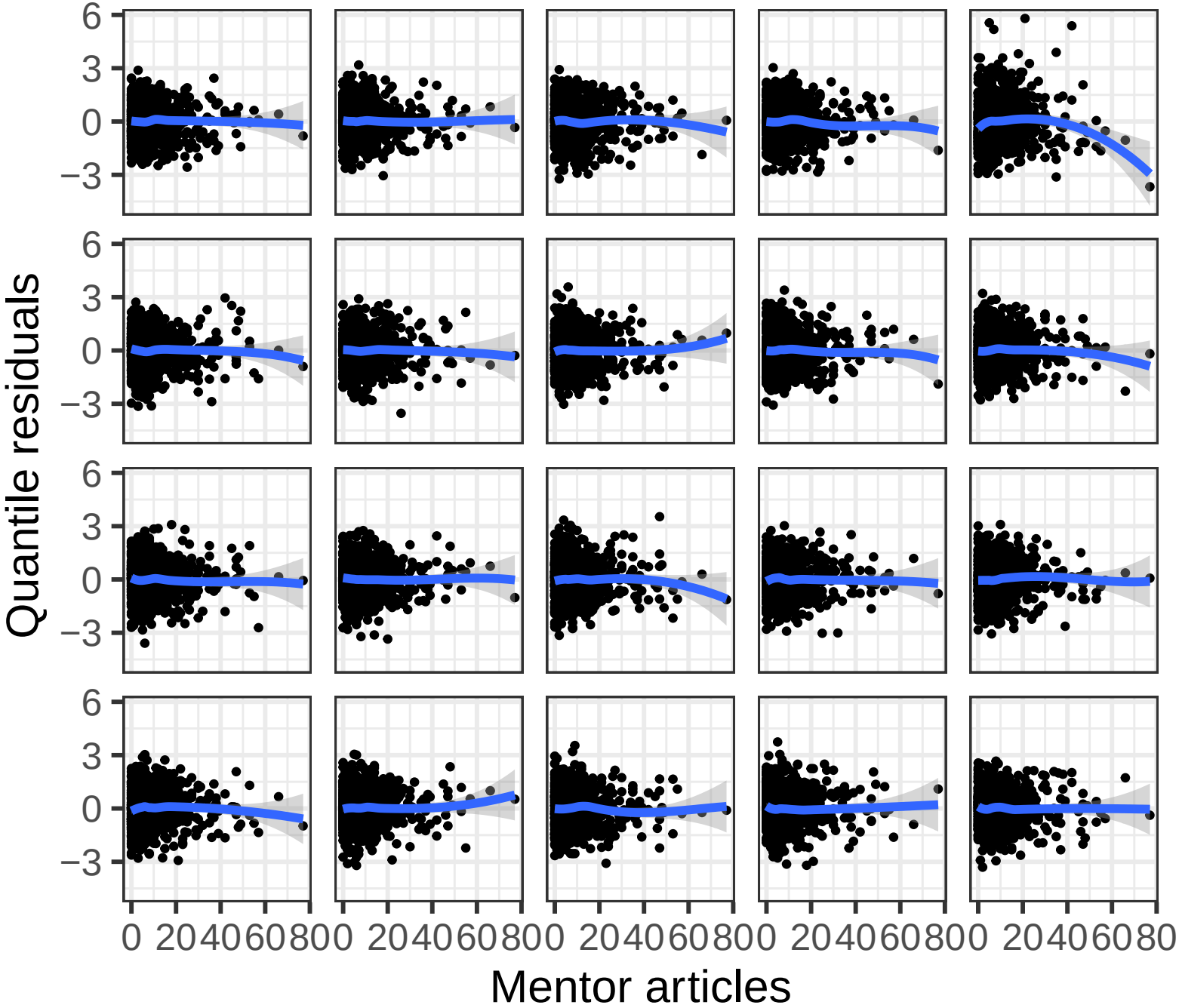
	art	phd	ment	kid5	cooks_distance
328	1	1.78	77	1	0.5587585
915	19	1.86	42	0	0.4113048
914	16	1.74	21	0	0.1478051
911	11	2.86	7	2	0.0905633
912	12	4.29	35	1	0.0638557
913	12	1.86	5	1	0.0625560
909	9	1.86	47	1	0.0481382
669	2	4.54	55	0	0.0351612
803	4	4.54	66	2	0.0243971
883	6	2.51	11	2	0.0199222

(Randomized) quantile residual plots



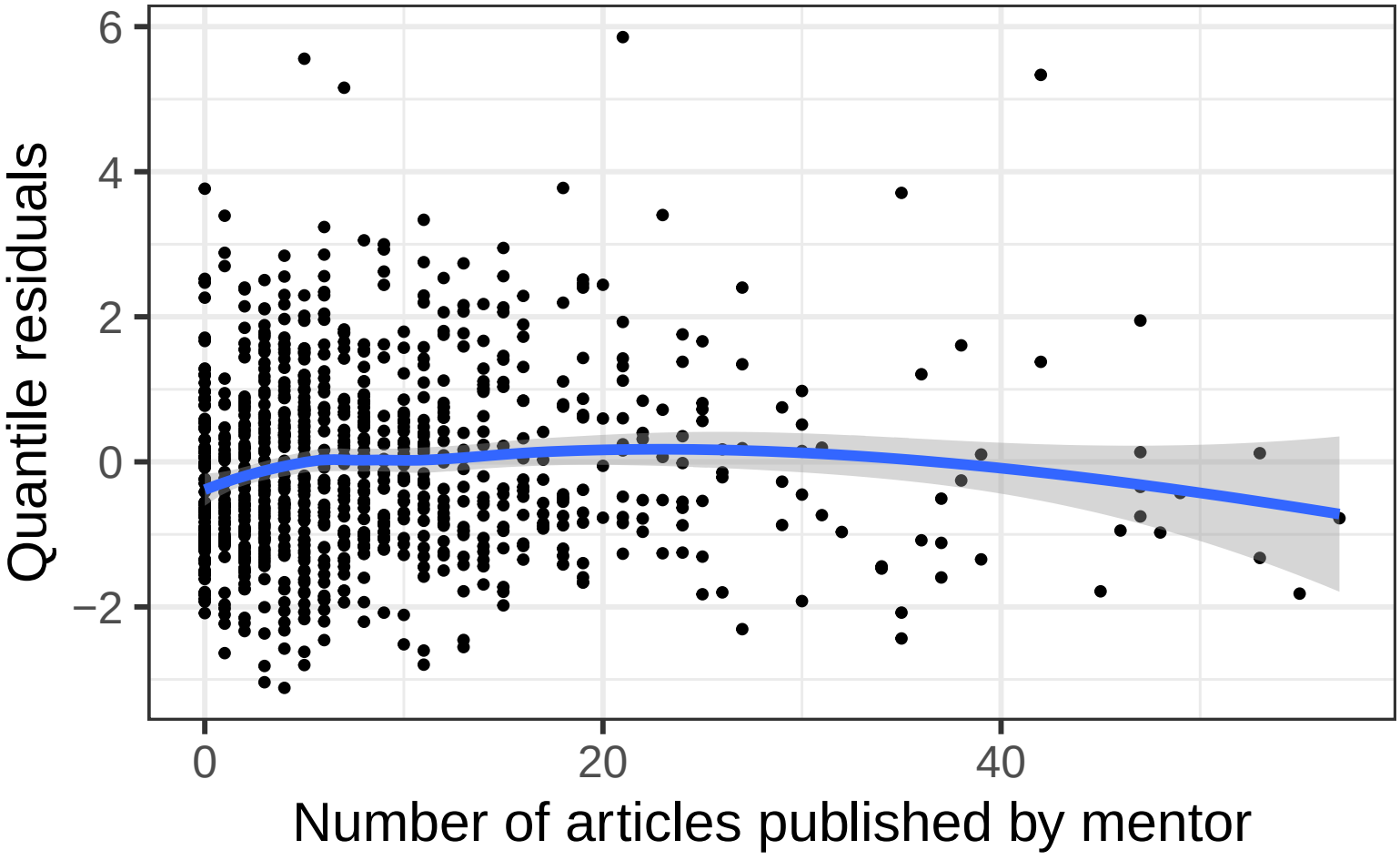
How could I decide whether deviations in the quantile residual plot are just due to random chance?

Plot lineup



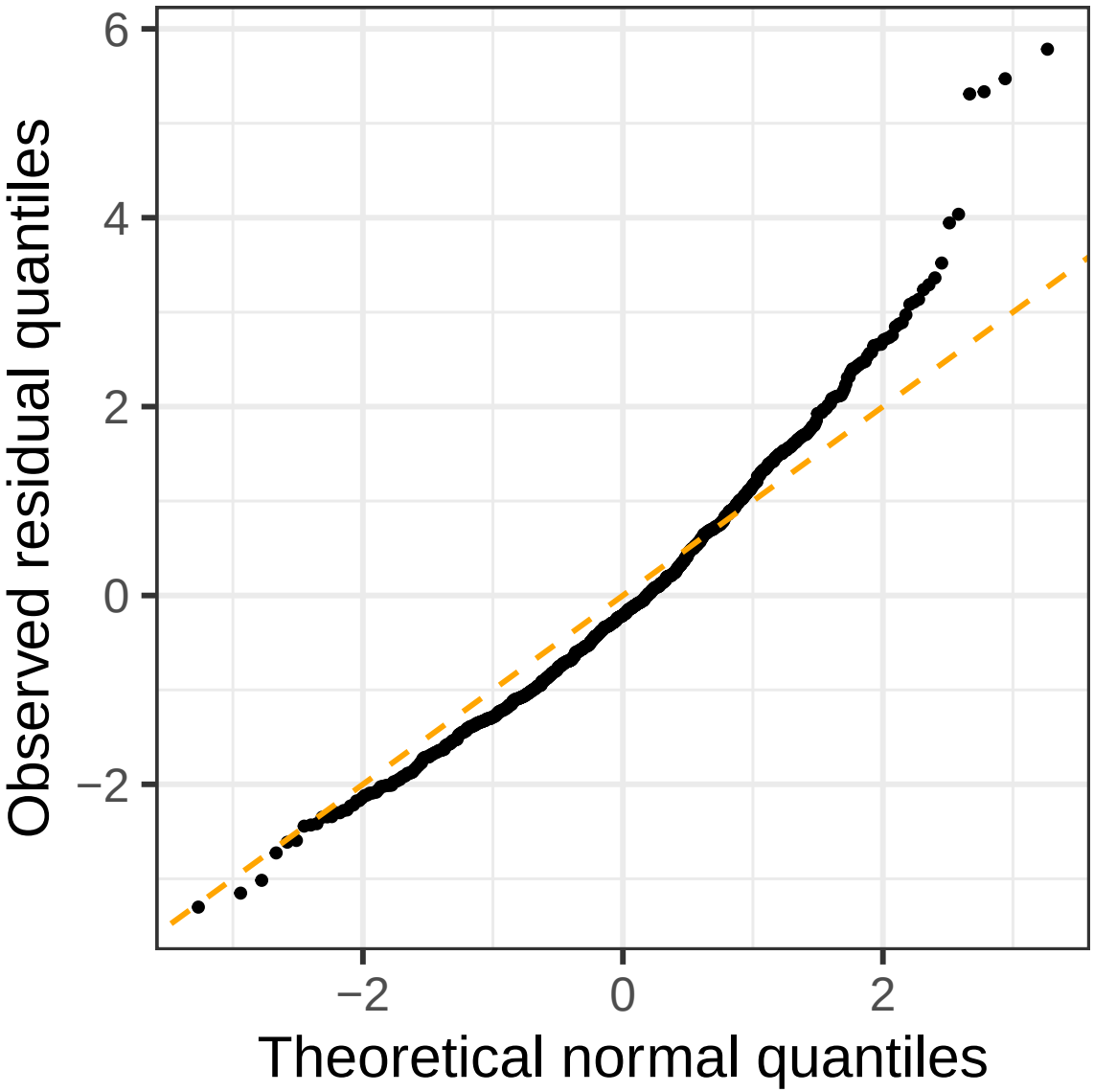
Quantile residual plot

Plot without the two rows with the most number of mentor publications:



QQ plot

If the model is a good fit to the data, randomized quantile residuals should be approximately $N(0, 1)$.



A goodness-of-fit test

A goodness-of-fit test

```
m_art <- glm(art ~ phd + ment + kid5,
             family = poisson, data = articles)

m_art$deviance
```

```
## [1] 1661.067
```

```
m_art$df.residual
```

```
## [1] 911
```

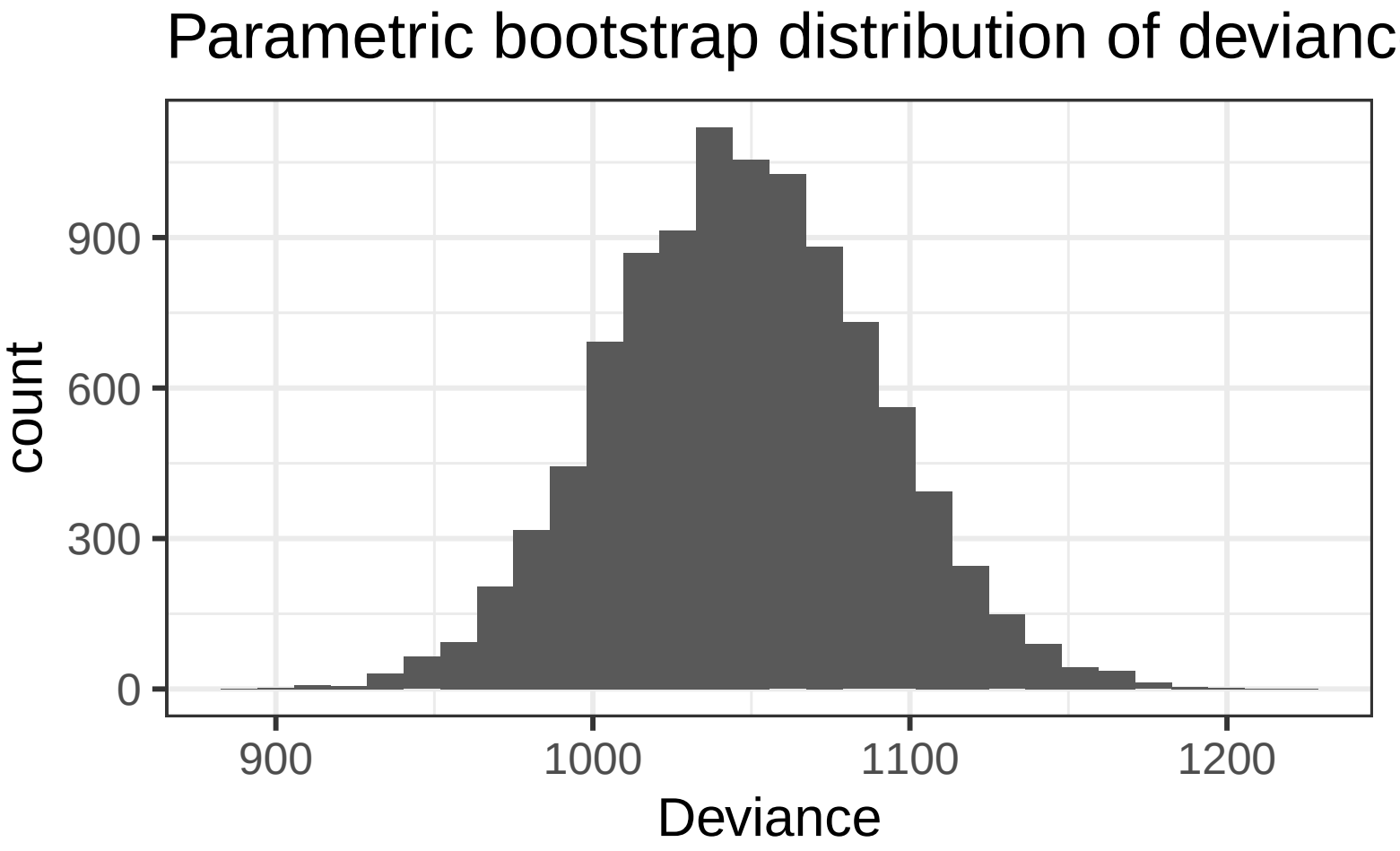
```
pchisq(m_art$deviance, m_art$df.residual, lower.tail=F)
```

```
## [1] 2.014849e-46
```

A goodness-of-fit test

Parametric bootstrapping

Parametric bootstrapping




```
m_art$deviance
```

```
## [1] 1661.067
```

Approximate null distribution

If the model is a good fit to the data, then the residual deviance approximately follows a χ^2 distribution, with the same degrees of freedom as the residual deviance.

```
m_art$deviance
```

```
## [1] 1661.067
```

```
m_art$df.residual
```

```
## [1] 911
```

Summary so far

For the Poisson regression on the articles data, here is my perspective:

- There are several potential outliers
- The rootogram shows our predicted counts are somewhat low in the tails of the distribution, compared to the observed counts
- Quantile residual plots suggest the shape assumption is overall pretty reasonable
- The QQ plot suggests that variability of the residuals is higher than expected
- The goodness-of-fit test suggests that some part of our model is mis-specified

Question: If this particular model is mis-specified, what do you think is the cause? How could we modify the model?